

Week 3 Summary

You should have learned about:

- cardinality and finite sets
- inclusion/exclusion
- The bit vector of a subset
- Ordering subsets by bit vector
- Multiple Cartesian Products
- Definition of strings of length n in an alphabet
- Lexicographical Order
- Finding the N 'th element in a multiple Cartesian product

Cardinality Facts:

(All sets are finite.)

$$\begin{aligned} |A^c| &= |\mathcal{U}| - |A| \\ \max(|A|, |B|) &\leq |A \cup B| \leq |A| + |B| \\ 0 &\leq |A \cap B| \leq \min(|A|, |B|) \end{aligned}$$

$$\begin{aligned} |\mathcal{P}(A)| &= 2^{|A|} \\ |\mathcal{P}_k(A)| &= \binom{|A|}{k} \\ |A \cup B| &= |A| + |B| - |A \cap B| \\ |A \cup B \cup C| &= |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C| \quad \text{inclusion/exclusion} \end{aligned}$$

Inclusion–Exclusion

The general relation: Given n sets, $A_1, A_2, \dots, A_n$.

$$\left| \bigcup_{k=1}^n A_k \right| = \sum_{\emptyset \neq I \subset \{1, 2, 3, \dots, n\}} (-1)^{|I|-1} \left| \bigcap_{i \in I} A_i \right|$$

$$\begin{aligned} |A_1 \cup A_2 \cup \dots \cup A_n| &= |A_1| + |A_2| + \dots + |A_n| + \\ &\quad - [|A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3| - \dots - |A_{n-1} \cap A_n|] \\ &\quad + [|A_1 \cap A_2 \cap A_3| - |A_1 \cap A_2 \cap A_4| - |A_3 \cap A_2 \cap A_4| - \dots - |A_{n-2} \cap A_{n-1} \cap A_n|] \\ &\quad (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n| \end{aligned}$$

Pascal's Triangle and k -subsets

If A is a set and $|A| = n$, then we know already that $|\mathcal{P}_k(A)| = \binom{n}{k}$.

Since

$$|\mathcal{P}(A)| = |\mathcal{P}_0(A)| + |\mathcal{P}_1(A)| + |\mathcal{P}_2(A)| + \cdots + |\mathcal{P}_n(A)|$$

we have the first of the two equations:

$$\begin{aligned} 2^n &= \sum_{k=0}^n \binom{n}{k} \\ 0 &= \sum_{k=0}^n (-1)^k \binom{n}{k} \end{aligned}$$

the second one coming from a golden argument.

Subsets, Bit Vectors and Binary Numbers

Given a set finite set A , with $|A| = n$ and a formal ordering on the elements of A ,

$$A = \{a_1, a_2, a_3, \dots, a_n\}, \quad a_i < a_j \text{ if } i < j$$

For any subset $B \subset A$, it's *bit vector* is a sequence of n 0's and 1's with the k 'th value (from the right) equal to 1 if $a_k \in B$, and 0 otherwise.

A bit vector of length k can be associated with a binary number with k digits, (zero's in front count.)

Example:

$\mathcal{P}(\{1, 2, 3\})$	$\emptyset :$	000
	$\{1\} :$	001
	$\{2\} :$	010
	$\{1, 2\} :$	011
	$\{3\} :$	100
	$\{1, 3\} :$	101
	$\{2, 3\} :$	110
	$\{1, 2, 3\} :$	111

Multiple Products

Let $A_1, \dots, A_N$ be sets.

$$A_1 \times A_2 \times \cdots \times A_N = \{(a_1, a_2, a_3, \dots, a_n) \mid a_i \in A_i\}$$

Example

$$A = \{0, 1, 2, 3, \dots, 9\} \quad B = \{a, b, c\}, \quad C = \{N, S, E, W\}$$

$$(c, (2, 2, 1), 1, S) \in B \times (A \times A \times A) \times A \times C = B \times A^3 \times A \times C$$

$$\begin{aligned} (b, (3, 2, 3, 2, 1), (1, 1, 1, 1, 2)) &\in B \times (A^5)^2 \\ (b, (3, 2), (3, 2), (1, 1), (1, 1), (1, 2)) &\in B \times (A^2)^5 \\ (b, (3, 2), (3, 2), (1, 1), (1, 1), (1, 2)) &\notin B \times (A^5)^2, \\ \text{So } (A^5)^2 &\neq (A^2)^5 \text{ in this notation.} \end{aligned}$$

Strings of Fixed Length

Set $\Sigma = \{a, b, c, d, e\}$.

The strings of length 8 on Σ correspond to Σ^8 and are usually written without parenthesis or commas.

aabaced versus (a, a, b, a, c, b, e, d)

The collection of all strings on Σ is

$$\Sigma^* = \bigcup_{k=0}^{\infty} \Sigma^k$$

Note 1: by definition, Σ^0 consists of just the *empty string*, $\Sigma^0 = \{\square\} = \{\lambda\}$

Note 2: $k \neq \infty$ so Σ^∞ is **not** in the union, so every string has *finite length*.

Lexicographic Order on A^N :

Assume A is a set, $A = \{a_1, a_2, \dots, a_{|A|}\}$, and the elements are ordered $a_1 < a_2 < \dots < a_{|A|}$.

Comparing $(a_1, a_2, \dots, a_N)$ with $(a'_1, a'_2, \dots, a'_N)$: if $a_1 = a'_1, \dots, a_{k-1} = a'_{k-1}$ and $a_k < a'_k$ then we say

$$(a_1, a_2, \dots, a_N) < (a'_1, a'_2, \dots, a'_N)$$

in lexicographic order (relative of the given order on the elements.)

Base N and lexicographic Order on A^N :

$|A^N| = |A|^N$, which is the same as the number of natural numbers with N digits in base $|A|$.

If you convert $k < |A|^N$ to base $|A|$, then k is represented as $d_{|A|-1}d_{|A|-2} \dots d_0$, that is,

$$k = d_{N-1}|A|^{N-1} + d_{N-2}|A|^{N-2} + \dots + d_0|A|^0$$

The k 'th element of A^N in lexicographic order (counting from 0) is

$$(a_{d_{N-1}}, a_{d_{N-2}}, \dots, a_{d_0})$$